

AI Code Reviewer — Design Document

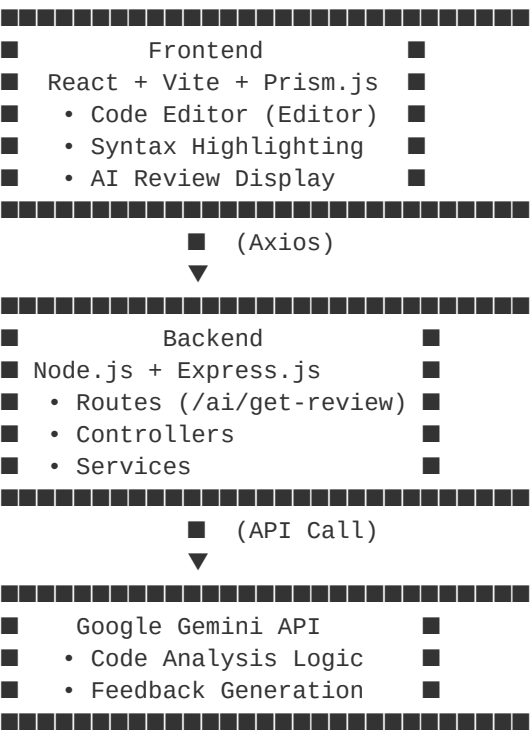
1. Overview

The AI Code Reviewer is a full-stack web application designed to analyze and review JavaScript code using the Google Gemini API. It provides intelligent feedback on code quality, readability, performance, and adherence to best practices. The system aims to assist developers by mimicking a senior code reviewer's approach through automated analysis and actionable suggestions.

2. Objectives

- Automate the code review process using Generative AI.
- Provide real-time feedback to developers on code structure, logic, and maintainability.
- Highlight potential improvements, errors, and optimization opportunities.
- Deliver an interactive UI for seamless code editing and review visualization.

3. System Architecture



4. Technology Stack

Layer	Technology	Purpose
Frontend	React (Vite), Prism.js, React Simple Code Editor, Marked.js, Prism.js	Interactive UI for code input and AI-generated review output
Backend	Node.js, Express.js, Axios	Handles API requests and integrates AI responses
AI Integration	Google Gemini API	Generates intelligent code reviews and improvement suggestions
Styling	CSS	Responsive and user-friendly layout

Communication	RESTful API	Enables seamless frontend-backend interaction
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5. Flow of Operation

- User enters or pastes code into the React editor.
- Frontend sends the code to the backend via Axios request.
- Backend routes the request to the AI controller, which calls the Gemini API.
- Gemini API analyzes the code and returns a structured review response.
- Backend sends the AI response to the frontend, which displays it clearly to the user.

6. Key Features

- AI-powered code analysis and review.
- Real-time feedback on best practices, logic, and maintainability.
- User-friendly and responsive UI built with React and Vite.
- Syntax highlighting and structured AI review results.
- Seamless API integration between frontend, backend, and AI services.

7. Conclusion

The AI Code Reviewer automates the code review process by leveraging the capabilities of the Google Gemini API and modern web technologies. It enables developers to improve their code quality, understand issues faster, and adopt better programming practices with minimal manual intervention.